

Lakshmi Sahasra Jangoan

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PROFESSIONAL SUMMARY

Data professional with 2.5+ years of experience across data analytics, data science, and data engineering, building cloud-ready data platforms, statistical/ML models, and BI dashboards from large, real-world datasets. Proficient in SQL, Python, and R for statistical modeling, predictive analytics, and time series forecasting (XGBoost, Random Forest, LSTM, ARIMA), and in Power BI, Tableau, and Excel for building dashboards and KPIs that drive stakeholder decisions. Experienced designing modern ETL/ELT pipelines with dbt, Airflow orchestration, and S3-compatible data lakes, and deploying analytics/ML services via FastAPI and Docker on Azure and PostgreSQL. Hands-on with deep learning (CNN/ResNet, LSTM/GRU) and LLM/RAG applications, with a proven ability to catch data quality issues early and communicate findings clearly to technical and non-technical stakeholders alike.

TECHNICAL SKILLS

Languages & Querying: Python (Pandas, NumPy, Scikit-learn, PyTorch), SQL (CTEs, Window Functions, Joins, Subqueries, Aggregations), R, MATLAB, JavaScript, Bash

BI, Analytics & Visualization: Power BI (DAX, Data Modeling), Tableau, Excel (Pivot Tables, VLOOKUP/XLOOKUP), Dashboard Development, KPI Reporting, Ad-hoc & Exploratory Data Analysis

Machine Learning & AI: Regression, Classification, Clustering (K-Means), XGBoost, Random Forest, Deep Learning (TensorFlow, PyTorch, CNN/ResNet, LSTM/GRU), Time Series Forecasting (ARIMA), LLMs & RAG, Hypothesis Testing, Cross-Validation

Data Engineering & Cloud: ETL/ELT Pipeline Design, dbt, Apache Airflow, MinIO (S3-Compatible Data Lake), PostgreSQL, Azure SQL, FastAPI, Docker, RESTful APIs, Data Validation & Quality Assurance, Git, GitHub, Pytest

PROFESSIONAL EXPERIENCE

Data Scientist - Research | *National Weather Center, University of Oklahoma* | *Norman, OK* 2024 – Jun 2026

- Built and compared 4 ML models (Random Forest, Logistic Regression, XGBoost, LSTM) combined with ResNet18-based deep learning feature extraction for precipitation forecasting, improving forecast accuracy by 14% over baseline.
- Processed 40+ years of ERA5 reanalysis data across 156 atmospheric channels, engineering spatiotemporal features and executed 600+ model training runs on HPC systems with PyTorch and Scikit-learn, reducing RMSE by 18% and MAE by 13%.
- Performed model diagnostics including calibration analysis and anomaly detection, translating results into visual summaries for cross-functional research stakeholders.

Graduate Teaching Assistant - Time Series Analysis | *University of Oklahoma* | *Norman, OK* Dec 2025 – May 2026

- Supported graduate-level instruction for 40+ students on ARIMA, exponential smoothing, seasonal decomposition, and predictive modeling using R, Python, Pandas, and statsmodels.
- Designed and graded assignments/exams and managed end-to-end course workflows for a graduate cohort.

Data Analyst | *ValueLabs* | *Hyderabad, India* Nov 2023 – Jul 2024

- Analyzed 120,000+ operational records using SQL and Python to drive KPI reporting and data-driven decisions across multiple departments, improving reporting accuracy by 15%.
- Built and deployed interactive Power BI dashboards and automated ETL workflows, reducing manual reporting effort by 25% and improving real-time visibility for senior stakeholders.
- Translated stakeholder reporting requirements into scalable analytical solutions, streamlining 10+ recurring reporting processes and structuring SQL data models for BI pipelines.

PROJECTS

Enterprise Urban Mobility Data Platform | *Python, PostgreSQL, dbt, Airflow, MinIO, FastAPI, Power BI, Docker*

- Built an end-to-end data platform processing 10.4M+ NYC taxi trip records, modeling fact/dimension tables in PostgreSQL and a dbt layer (10 models, 40 passing tests) orchestrated via an 8-task Airflow DAG with an S3-compatible MinIO data lake.
- Exposed analytics and ML prediction endpoints via FastAPI, built a 5-page Power BI dashboard, and trained a trip-duration model achieving 0.90 R^2 (1.47 min MAE), validated with 11 Pytest tests and zero data-quality violations.

Healthcare Time Series Forecasting for Public Health Monitoring (Peru) | *Python, ARIMA, XGBoost, LSTM, Tableau*

- Contributed to an OU-UNSA international public health research grant, building a national-level forecasting pipeline on 20+ years of epidemiological surveillance data, comparing 7 models with rolling-window validation; XGBoost achieved the best results (R^2 of 0.935 for children, 0.890 for adults).
- Engineered lag/rolling/seasonal features, generated confidence intervals for forecast uncertainty, and built an interactive Tableau dashboard for public health surveillance and outbreak monitoring.

Local AI Assistant with RAG Pipeline | *Python, Ollama, FastAPI, ChromaDB, LLMs*

- Built a fully local retrieval-augmented generation (RAG) pipeline benchmarking 3 open-source LLMs, reaching 45 tokens/sec (2.4x faster, 53% less memory than Mistral 7B) with 100% source retrieval accuracy.
- Engineered structured JSON output using Pydantic validation (80% schema reliability) and built a Streamlit UI with live token/latency metrics and document-based RAG chat.

National Park Relational Database System | *SQL, Azure SQL, Java, ER Modeling*

- Designed a normalized (3NF) relational schema with ER diagrams spanning 25+ entities, applying B+ tree and clustered indexing in Azure SQL to improve query efficiency.
- Built 15 SQL stored procedures/workflows and a Java JDBC application (18 operations) supporting end-to-end CRUD, reporting, and transaction management with full import/export validation.

EDUCATION

University of Oklahoma (GPA: 4.0/4.0)

Master of Science, Data Science and Analytics

Aug 2024 – May 2026

Norman, OK

Jawaharlal Nehru Technological University

Bachelor of Technology, Computer Science and Engineering

Dec 2020 – May 2024

Hyderabad, India